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Lab 5 - Part B: Open Jackson Queueing Network
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Input parameters:

```
lambda1 = 10.0000000000 clients/sec
lambda2 = 6.0000000000 clients/sec
mu1      = 15.0000000000 clients/sec
mu2      = 20.0000000000 clients/sec
mu3      = 10.0000000000 clients/sec
mu4      = 14.0000000000 clients/sec
mu5      = 12.0000000000 clients/sec
```

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B.2 - Effective arrival rates and traffic intensities
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Effective arrival rates Γ_i :

```
Gamma1 = 10.0000000000 clients/sec
Gamma2 = 11.0000000000 clients/sec
Gamma3 = 6.1666666667 clients/sec
Gamma4 = 7.3333333333 clients/sec
Gamma5 = 10.5666666667 clients/sec
```

Traffic intensities $\rho_i = \Gamma_i / \mu_i$:

```
rho1 = 0.6666666667
rho2 = 0.5500000000
rho3 = 0.6166666667
rho4 = 0.5238095238
rho5 = 0.8805555556
```

Ergodicity flag: 1

The network is ergodic because all ρ_i are smaller than 1.

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B.3 - Mean number of clients in each queue
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Mean clients $E[N_i]$ in each queue:

```
E[N1] = 2.0000000000
E[N2] = 1.2222222222
E[N3] = 1.6086956522
E[N4] = 1.1000000000
E[N5] = 7.3720930233
```

Total mean number of clients in the network:

```
E[N] = sum_i E[N_i] = 13.3030108977
```

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B.4 - End-to-end mean delay for the given parameters
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Total external arrival rate:

```
gamma_external = lambda1 + lambda2 = 16.0000000000 clients/sec
```

End-to-end mean delay by Little's Law:

```
E[T] = E[N] / gamma_external = 0.8314381811 seconds
```

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B.5 - Bottleneck queue and maximum lambda1
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Current bottleneck queue, based on maximum ρ_i :

```
bottleneck = Q5
```

max rho = 0.8805555556

Upper bounds on lambda1 imposed by each queue:

From Q1: lambda1 < 15.0000000000
From Q2: lambda1 < 28.0000000000
From Q3: lambda1 < 19.2000000000
From Q4: lambda1 < 30.0000000000
From Q5: lambda1 < 12.3243243243

Maximum lambda1 for ergodicity with lambda2 fixed at 6.0000000000:

lambda1_max = 12.3243243243 clients/sec

limiting queue = Q5

Current lambda1 = 10.0000000000 is below lambda1_max, so the current network is ergodic.

B.6 - End-to-end delay plot versus lambda1

Plot range:

lambda1_min = 1.2324324324 clients/sec
lambda1_max_plot = 12.2010810811 clients/sec
lambda1_max_ergodic = 12.3243243243 clients/sec

Delay values at selected points:

lambda1 = 0.10 * lambda1_max = 1.2324324324 -> E[T] = 0.2950551294 sec
lambda1 = 0.25 * lambda1_max = 3.0810810811 -> E[T] = 0.3323554951 sec
lambda1 = 0.50 * lambda1_max = 6.1621621622 -> E[T] = 0.4349166146 sec
lambda1 = 0.75 * lambda1_max = 9.2432432432 -> E[T] = 0.6880326518 sec
lambda1 = 0.90 * lambda1_max = 11.0918918919 -> E[T] = 1.2973780957 sec
lambda1 = 0.95 * lambda1_max = 11.7081081081 -> E[T] = 2.2074842154 sec
lambda1 = 0.99 * lambda1_max = 12.2010810811 -> E[T] = 9.1519048705 sec

Saved figure as lab5_part2_delay.png

FINAL SUMMARY - Lab 5, Part B

Routing equations:

Gamma1 = lambda1
Gamma2 = lambda2 + (1/2)lambda1
Gamma3 = (5/12)lambda1 + (1/3)lambda2
Gamma4 = (1/3)lambda1 + (2/3)lambda2
Gamma5 = (37/60)lambda1 + (11/15)lambda2

Given parameters:

lambda1 = 10.0000000000, lambda2 = 6.0000000000
mu = [15.0000000000 20.0000000000 10.0000000000 14.0000000000 12.0000000000]

Effective arrival rates:

Gamma = [10.0000000000 11.0000000000 6.1666666667 7.3333333333
10.5666666667]

Traffic intensities:

rho = [0.6666666667 0.5500000000 0.6166666667 0.5238095238 0.8805555556]
ergodic = 1

Mean clients:

E[N_i] = [2.0000000000 1.2222222222 1.6086956522 1.1000000000 7.3720930233]
E[N] = 13.3030108977

End-to-end delay:

E[T] = 0.8314381811 seconds

Bottleneck:
current bottleneck = Q5, $\rho = 0.8805555556$
 $\lambda_{\max} = 12.3243243243$ clients/sec
limiting queue for $\lambda_{\max} = Q5$